PHOTOVOLTAIC ARRAY

-GP1

STRING 1

STRING 2

STRING 3

/ / / STRINGS 4 TO 8 AS ABOVE

136 No. JINKO JKM580N 580 W
78.88 kWp, 8 STRINGS OF 17 IN SERIES
Voc 956 V AT -5 C, Isc 14.01 A PER STRING

8 SEPARATE STRING CIRCUITS, 1 PER MPPT
NOT PARALLELED, DC SWITCH INTEGRAL TO -TC1

8 CIRCUITS -W1

-FV1
DC SPD
TYPE 2, 1000 V, 1 PER MPPT, ARRAY END

ARRAY FRAME BOND 16 mm2

BATTERY SYSTEM

-GB1

9 No. FOXESS CQ7, LFP
63.18 kWh, 518.4 V, 41.5 kW
56.86 kWh USABLE, 0.66 C

-QS2 BATTERY ISOLATOR
1000 V, 125 A

-QF1 BATTERY MCB
125 A, 2 POLE, 1000 V DC

2 x 1C -W2

-KB1
BMS AND EPO LOOP
CAN TO -TC1, EPO AT DOOR

CAN BUS, CHARGE LIMITS

16 mm2 PE

MAIN EARTH BAR

-TC1
FOXESS P3-80-Plus
80 kW HYBRID, 400 V 3PH

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-TC1 INTEGRAL FUNCTIONS

8 MPP TRACKERS, 1 STRING EACH
INTEGRAL DC SWITCH
RCMU TYPE B
ANTI ISLANDING TO IEC 62116
3 BATTERY PORTS, 1 IN USE
DC AND AC SPD, TYPE 2

25 mm2 PE

-QS3 AC ISOLATOR
250 A 4 POLE

-FV2
AC SPD
TYPE 2, 4 POLE

-TA1
200/5 A

-W3
4 CORE

-QF4 EPS MCB
63 A, 4 POLE

-W4

-EPS1
ESSENTIAL SERVICES BOARD

4 OUTGOING WAYS, 15 kW TOTAL
FED AT ALL TIMES FROM -TC1
TRANSFER INTERNAL, UNDER 10 ms
NO LINK TO THE LV BOARD

AC CONNECTION AND METERING

-P1
GENERATION METER
kWh

-KF1
INTERFACE PROTECTION
27 / 59
81 / ROCOF

TRIP TO -QF2

-T1
UTILITY TRANSFORMER, BY OTHERS
11 kV / 400 V

POINT OF SUPPLY

EXISTING MAIN LV BOARD

-TA2
800/5 A, EXPORT LIMIT

-P2
IMPORT EXPORT METER
kWh

EXPORT SIGNAL TO -TC1

-QF0
630 A INCOMER

-QF2
NEW WAY
160 A 4 POLE, SHUNT TRIP

400 V 3PH + N BUSBAR, 630 A, 25 kA

-QF5

-QF6

EXISTING SITE LOAD
85 kW PEAK, ASSUMED PROFILE

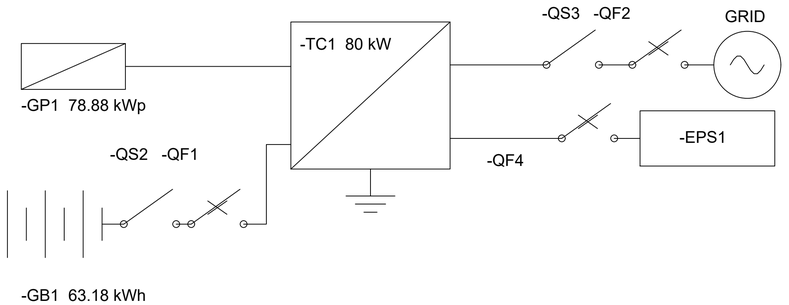
MET AND N-E LINK

-W5

MAIN EARTHING CONDUCTOR 50 mm2

ELECTRODE RESISTANCE UNDER 10 OHM, TN-S

KEY DIAGRAM



LEGEND

SWITCH DISCONNECTOR

CIRCUIT BREAKER

SURGE ARRESTER, TYPE 2

CURRENT TRANSFORMER

ENERGY METER (kWh)

PROTECTION RELAY (27/59)

DISTRIBUTION TRANSFORMER

BATTERY, SERIES STRING

PHOTOVOLTAIC STRING

UTILITY SUPPLY

PROTECTIVE EARTH

CABLE SCHEDULE

TAG	CABLE	LENGTH	PROTECTION
-W1	16 x 1C 6 mm2 PV1-F	45 m	NONE, 1 PER MPPT
-W2	2 x 1C 35 mm2 Cu XLPE	8 m	-QF1 125 A DC
-W3	4C 95 mm2 Cu SWA	32 m	-QF2 160 A
-W4	4C 35 mm2 Cu	12 m	-QF4 63 A
-W5	1C 50 mm2 G/Y	20 m	EARTHING

SIZES AND VOLT DROP TO BE CONFIRMED ON ROUTE

INTERFACE PROTECTION SETTINGS, TO BE CONFIRMED BY IESCO

FUNCTION	SETTING	TRIP TIME
27 UNDER VOLTAGE	0.80 Un	2.5 s
27 STAGE 2	0.50 Un	0.5 s
59 OVER VOLTAGE	1.15 Un	1.0 s
81 UNDER FREQUENCY	47.5 Hz	0.5 s
81 OVER FREQUENCY	51.5 Hz	0.5 s
ROCOF	1.0 Hz/s	0.5 s
RECONNECT DELAY		180 s

NOTES

1 NO STRING FUSES. ONE STRING PER MPP TRACKER, SO NO PARALLEL FAULT PATH EXISTS.
2 BATTERY MODULES IN SERIES ONLY. NO PARALLEL STRINGS, ONE CURRENT PATH.
3 -QF1 IS ABOVE THE 110 A SIXTY SECOND PEAK AND BELOW THE 35 mm2 CABLE RATING.
4 SoC HELD BETWEEN 10 AND 100 %. 56.86 kWh USABLE, 3.79 h BACKUP FROM FULL.
5 -EPS1 IS FED ONLY FROM THE -TC1 EPS PORT. ITS GRID BYPASS IS INTERNAL TO -TC1.
6 EARTHING TN-S. ARRAY FRAME, BATTERY ENCLOSURE AND -TC1 BONDED TO THE EARTH BAR.
7 EXPORT LIMITED BY SIGNAL FROM -TA2. CHECK SETTING AT COMMISSIONING.

REV	DESCRIPTION	DATE	BY	CHK
P1	FIRST ISSUE FOR COMMENT	12.09.2026	UJ	--
P2	SPD AND METERING ADDED	28.09.2026	UJ	--
P3	ISSUED FOR COMMENT	08.10.2026	UJ	--

CLIENT
SAMPLE COMMERCIAL CLIENT

PROJECT
80 kW ROOFTOP PV WITH 63 kWh BATTERY STORAGE
I-10 INDUSTRIAL AREA, ISLAMABAD

DRAWING TITLE
SINGLE LINE DIAGRAM

DRAWING No.
PV-04

DRAWN USAMA JAVED CHK -- APP --

SCALE NTS @ A1 REV P3